

Reduced vigor of a clonal invader: Lack of evidence for soilborne pathogens



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ABSTRACT

Predicting the impacts of exotic species on native ecosystems requires an understanding of the ecological processes that regulate biological invasions. Despite accumulating evidence that belowground interactions play a significant role in determining the outcome of plant invasions, few studies have examined the role of soilborne pathogens in invasion dynamics. We tested the hypothesis that accumulating negative plant–soil feedbacks reduce the vigor of clones of smooth brome (*Bromus inermis*), invading northern fescue prairies in Manitoba, Canada. We examined the density, biomass, and height of ramets of smooth brome from the center of invading clones to their fringes and tested the negative plant–soil feedback hypothesis using plant parasitic nematodes and soilborne pathogens. Ramets of smooth brome were less vigorous inside the invading clones compared with those along the fringes. The observed loss of vigor inside the clones was not related to plant parasitic nematodes or soilborne pathogens. Fewer soilborne plant parasitic nematodes were found at the center of invading clones and pasteurization had no effect on the mass of seedlings raised in soil collected in this area. Higher available soil nitrogen, phosphorus, and soil moisture at the center of invading clones suggest senescence in the older areas of the clone. However, other hypotheses, including potential impacts of other groups of plant pathogens and autotoxic effects from the accumulating litter, remain unexplored. Despite the reduced vigor of smooth brome at the center of invading clones, plant communities in this area remained less diverse. Future restoration of native prairies invaded by smooth brome will require active methods to reverse these negative impacts.

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1. Introduction

Control of exotic organisms requires an understanding of both their impacts on native ecosystems and the ecological processes that regulate their invasions. Accumulating evidence that biotic interactions belowground play a significant role in determining the outcome of plant interactions (Bever, 2003) promises important insights into mechanisms that determine the success of biological invaders. Among complex, multi-trophic interactions between plants and soilborne organisms, soilborne pathogens remain a poorly studied element of invasion dynamics (Eppinga et al., 2006).

Pathogens affect host populations and communities of plants through their impacts on the survival, growth and fecundity of individuals (Gilbert, 2002). Losses of carbon and nutrients to enemies result in lower growth rates and decreased primary

productivity (Mitchell, 2003). Negative feedbacks between plants and soilborne pathogens also help maintain local community diversity by preventing individual species from increasing in dominance (Bonanomi et al., 2005; Mangan et al., 2010). Negative feedbacks are favored where both pathogens and hosts have limited dispersal (Gilbert, 2002). For example, at the interface of plant roots and soil, host-specific microorganisms, including root feeding nematodes and pathogenic fungi, may accumulate over time, increasing their biological effects on their host (Bever, 2003).

Declines in the vigor of monospecific stands of perennial plants are well documented in both natural and cultivated ecosystems. Along coastal dunes, reductions in the height and density of beachgrass (*Ammophila arenaria* (L.) Link) have been attributed to the gradual accumulation of pathogenic nematodes (de Rooij-van der Goes, 1995). Nematodes and pathogenic fungi are also responsible for declines in the vigor of red fescue (*Festuca rubra* L.) and sand sedge (*Carex arenaria* L.) in the Netherlands (Oloff et al., 2000) and the successional dynamics of foredune and temperate forest vegetation (van der Putten et al., 1993; Packer and Clay,

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2003). Perennial crops are also susceptible to accumulating populations of pathogens. For example, the management of crop residues and soil microbial communities remains key to the suppression of the take-all patch fungus (*Gaeumannomyces graminis* (Sacc.) Arx & D. Oliver) and other soilborne pathogens (Cox et al., 2005).

The strength of the feedback between plants and soilborne pathogens is often a function of the age, the size, and the density of host populations (Carlsson et al., 1990; Packer and Clay, 2003). In crop monocultures, cumulative losses in plant productivity are often linked to qualitative and quantitative changes in communities of pathogens (Shipton, 1977). Often, yields can only be recovered by rotating crops and interrupting cycles of disease (Krupinsky et al., 2004). In native communities, negative feedbacks between soilborne pathogens and plants can also increase species turnover and maintain plant biodiversity (Petermann et al., 2008). For example, the observed disproportional mortality of seedlings near parent plants forms the basis of the escape hypothesis, which proposes a selective advantage of dispersal and predicts increased rates of infection among common species (Packer and Clay, 2003).

The link between disease incidence and the spatial structure of populations poses a unique problem for plant species forming dense, perennial stands, especially those that grow clonally. Compared with annual species, clonal perennial grasses are often associated with a greater number of fungal pathogens (Clay, 1995) and the observed central die-back among clonal species can be induced by the build-up of negative plant–soil feedback in the central, older areas of expanding clones (Carteni et al., 2012; Bonanomi et al., 2014). Higher parasite pressures are also been reported among perennial species in the family of Caryophyllaceae (Thrall et al., 1993) and larger hosts, such as shrubs and trees (Strong and Levin, 1979). Although exotic plants may have escaped native enemies, their long-term persistence may still depend on the build-up of pathogens (Dietz et al., 2010). For example, some exotic invaders do not maintain their invasive rank (Perrins et al., 1993) and gradually disappear from their introduced environments (Crawley, 1987). Especially vulnerable may be exotics that rapidly displace native species to form dense monocultures (Carlsson et al., 1990). Among the large number of exotic species in most regional floras, the ability to grow clonally, shared by the majority of successful invaders (Pyšek, 1997) might also be expected to contribute to their vulnerability to plant pathogens. For example, given that clonal growth can result in substantial lateral spread, large clones may provide bigger targets for pathogen colonization than smaller, individual plants (Clay and van der Putten, 1999).

In this study, we examine the role of soilborne pathogens in the decline of smooth brome (*Bromus inermis* Leyss.), a Eurasian perennial, invading native prairies throughout the Great Plains (Otfinowski et al., 2007). Widely planted for hay, forage, and to revegetate disturbed areas, smooth brome often escapes to establish in areas of native prairie, spreading clonally to produce dense monocultures that compromise the structure and function of invaded communities (Bennett et al., 2014). Despite its susceptibility to a high number of fungal pathogens (Clay, 1995), including species of the root pathogens, *Fusarium* and *Pythium*, and the root-lesion nematode *Pratylenchus penetrans* (Cobb) Filipjev & Schuurmans-Stekhoven (Otfinowski et al., 2007), existing research has not explored questions regarding the long-term persistence of smooth brome in native communities. The aim of our study was to examine the vigor of smooth brome clones invading native prairies. Using a combination of field surveys and greenhouse experiments, we examined the vigor of invading clones, from their centers to their invading fringes, and tested the hypothesis that accumulating negative plant–soil feedbacks at the center of invading clones will reduce their vigor.

2. Materials and methods

2.1. Study area

Research was conducted in Riding Mountain National Park, Manitoba, Canada (latitude: 50.6578, longitude: –99.9720). The park occupies an area of 2978 km² in western Canada and consists of large areas of rolling upland (550–640 m), underlain by glacial tills (Cody, 1988). The region is characterized by a mean annual precipitation of 475 mm, mean temperatures between –18 °C (January) and 18 °C (July), and a growing season of 170 days (Leeson et al., 2005). Riding Mountain National Park lies in the Mixedwood Section of the Boreal Forest Region and protects areas of northern fescue prairie (Cody, 1988). In North America, fescue prairies form an arc around the northern and northwestern perimeter of the mixed prairie, occupying a transition between the warmer and drier grasslands and the cooler, moister aspen parkland (Coupland and Brayshaw, 1953). Fescue prairies, dominated by rough fescue (*Festuca hallii* (Vasey) Piper) and species of *Elymus* and *Carex*, are threatened in western Canada by the impacts of grazing, cultivation and exotic species invasions (Trottier, 1986).

2.2. Surveys of invading clones

We randomly selected 34 clones of smooth brome invading two fescue prairies (Bob Hill and Strathclair) in Riding Mountain National Park (17 clones per prairie). Selected clones ranged from 120 to 1370 cm in diameter. The two prairies were situated approximately 50 km apart; soils at the more western Bob Hill prairie (latitude: 50.9187; longitude: –100.8865) were more xeric and lower in mineral nitrogen than the centrally located Strathclair prairie (latitude: 50.7932; longitude: –100.2401).

Based on a factorial design, we divided each clone into three areas (center, middle, and fringe) and used 50 × 50 cm quadrats, placed along transects (bearing: 135°), to measure the density, height, and biomass of brome ramets and to collect samples of root biomass, thatch and litter. Thatch and litter, defined as standing, fallen, and variously disintegrated residues of plants (Agriculture Canada, 1998), were not distinguished by species. During July 2005, ramets of smooth brome were clipped at the soil surface inside each quadrat, air dried (48 h, 60 °C), and weighed. We collected samples of root biomass from a subset of six clones per prairie ($n = 12$). Root cores were collected inside each quadrat using a hand auger (diameter: 5 cm) and separated by depth (0–10 cm, 10–30 cm). Cores were cleaned using a hydropneumatic root washer (Gillison's Variety Fabrication, Inc., Benzonia, MI), air dried (48 h, 60 °C), and weighed. No distinction was made between the roots of smooth brome and other species. During August 2004, we measured the vegetative cover of vascular plants at the center, middle, and along the fringes of invading clones using 1 × 1 m quadrats.

Quadrats at the center, middle, and fringes of invading clones were also used to collect soil samples for nitrogen and phosphorus extraction. In addition, we used in situ incubations to monitor changes in mineral nitrogen in all 34 clones (Binkley and Hart, 1989). Cores of soil, extracted using a hand auger (0–15 cm, 74 cm³), were placed in plastic Ziploc[®] bags, buried in the auger excavations, and incubated between July and August 2004. Soil incubations in plastic bags allow the exchange of oxygen and carbon dioxide (Gordon et al., 1987). Along with reference samples, collected at the time of installation, all soil samples were air dried, milled, extracted with 2.0 M KCl, and analyzed for ammonium and nitrate using a Technicon[®] AutoAnalyzer[®] II (Maynard and Kalra, 1993). Due to rodent damage, only results from undamaged incubation bags at one prairie are presented here (Strathclair, $n = 16$). We repeated our measurements of soil mineral nitrogen

during July 2005. Samples of air-dried soil collected in 2005 were extracted with 0.5 M NaHCO_3 and analyzed for available phosphorus (Olsen et al., 1954).

2.3. Impacts of plant parasitic nematodes and soilborne pathogens

We tested the hypothesis that accumulating negative plant–soil feedbacks reduce the vigor of invading clones using plant parasitic nematodes and soilborne pathogens. Samples of nematodes were extracted from soil collected from the center, middle, and along the fringes of six clones invading the two study prairies ($n = 12$). Selected clones were characterized by decreased vigor of smooth brome ramets at their center. At each clone, we excavated samples of soil (0–15 cm, $\sim 1200 \text{ cm}^3$ composite of four cores) around four ramets of smooth brome at the center, middle, and along the fringe of invading clones. Samples were collected during August 2004, stored at 5°C , and analyzed within three weeks.

Extractions of soilborne nematodes were based on the Cobb sieving and flotation method (Kimpinski, 1993). Nematodes were extracted at room temperature from 100 g of field soil (400-mesh size; mean soil moisture: 4.8%), centrifuged ($420 \times g$, 5 min), suspended using a 1 M sucrose solution (specific gravity 1.18), centrifuged ($420 \times g$, 60 s) and sieved (500-mesh sieve), counted, and identified. Identification of plant parasitic nematodes from soil extracts followed Mai and Mullin (1996) and Yeates et al. (1993).

The impact of soilborne pathogens on the vigor of smooth brome seedlings was tested in the greenhouse. We collected samples of soil (0–15 cm, $\sim 900 \text{ cm}^3$ composite of three cores) in the center, middle, and along the fringes of the twelve clones used in the extraction of plant parasitic nematodes. Soil samples were cleaned of rocks and any roots contained in the soil were cut into

1 cm pieces. We split each soil sample in half and pasteurized one portion. Samples were pasteurized using an automatic soil pasteurizer ($<99^\circ\text{C}$, 1 h; Dillion Industries Inc., Melrose, MA). Non-pasteurized and pasteurized soils were potted into eight pots (diameter = 11 cm) that were surface sterilized with 70% ethanol and lined with paper towels. All pots were at least half full ($\sim 450 \text{ cm}^3$), but we did not specifically control for the weight and volume of soil. We left potted soil idle for one week, watering it daily, to allow the free drainage of nutrients mineralized during pasteurization. Due to space limitations, we used soil from six of the twelve clones in the greenhouse experiment.

Seedlings of smooth brome used in the greenhouse experiment were germinated from seeds collected at each study site during October 2004. Collected seeds were stored in the dark for two months (5°C) and germinated in a growth chamber ($25^\circ\text{C}/15^\circ\text{C}$, 12/12 h light/dark, $107 \mu\text{mol m}^{-2} \text{s}^{-1}$) on plastic Petri dishes (9 cm), lined with filter paper. Seeds were not surface sterilized and were watered with de-ionized water, as required. Using tweezers, we transplanted three seedlings with emerging cotyledons into single pots of pasteurized or unpasteurized soil from the same prairie site as the collected seeds. Greenhouse conditions were maintained between 20 and 25°C (16/8 h) and natural light was supplemented with sodium lamps. Between December 2004 and July 2005, we raised five consecutive generations of brome seedlings in each pot and harvested at the end of each five-week cycle. We based this experimental period on observation of differences in the vigor of seedlings at the end of each growth cycle. We fertilized pots of unpasteurized soil with 20–20–20 N–P–K fertilizer (53.26 mg N/kg soil) during the third week of the second growth cycle to compensate for nitrogen mineralized during soil pasteurization, but reduced this rate by half during the third growth cycle. At the end of each five-week growth cycle, we clipped all seedlings at the

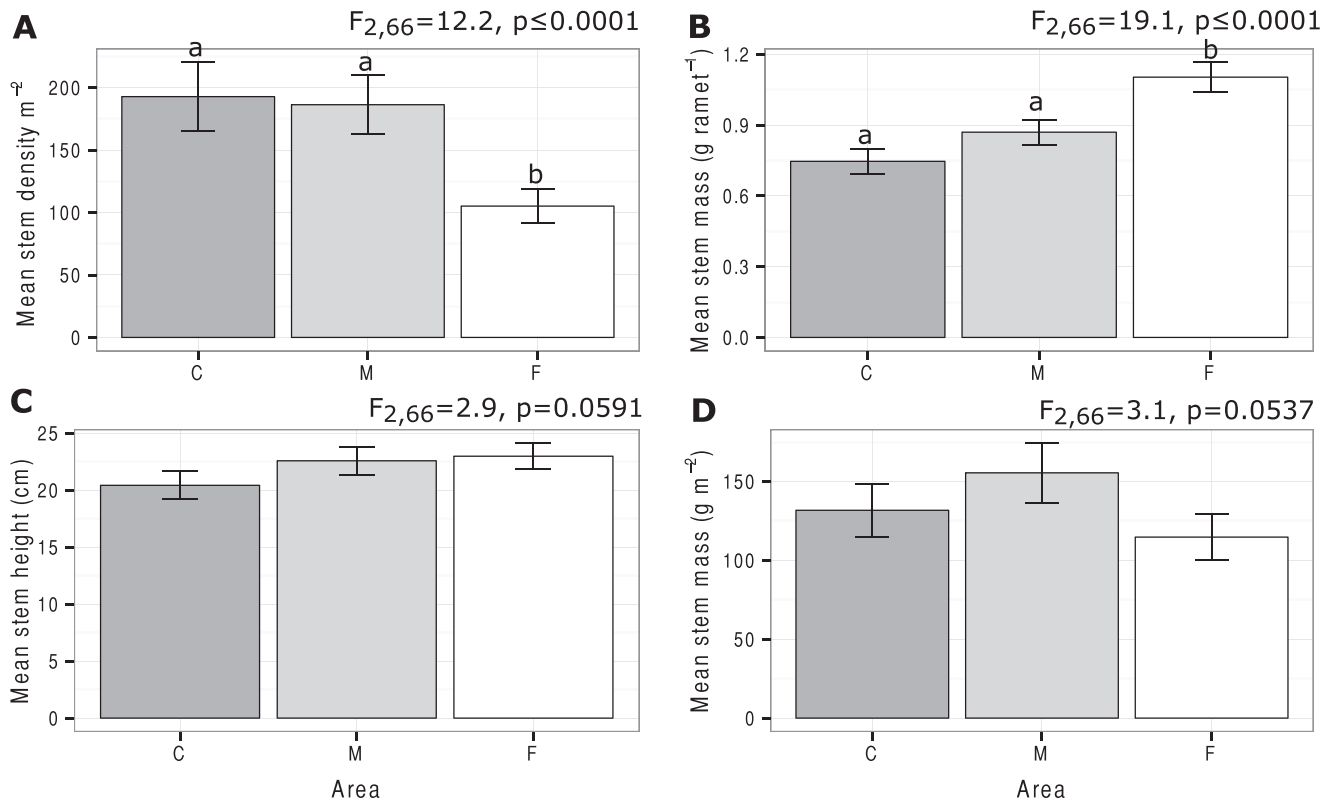


Fig. 1. Summary of the density (m^{-2}), aboveground biomass (g ramet^{-1} , g m^{-2}), and height (cm) of smooth brome (*Bromus inermis* Leyss.) invading two northern fescue prairies in Riding Mountain National Park, Manitoba, Canada. Data were collected inside 50×50 cm quadrats, placed at the center (C), middle (M), and along the fringe (F) of invading clones. Means of treatment groups were blocked by clone ($n = 34$) to reduce variability between individual clones within the one-way ANOVA. Means ± 1 S.E.

Table 1

Plant community structure and environments across clones of smooth brome (*Bromus inermis* Leyss.) invading two northern fescue prairies in Riding Mountain National Park, Manitoba, Canada. The areas center, middle, and fringe, represent a transect from the center of invading clones to their prairie fringes. Means of treatment groups were blocked by clone ($n = 34$) to reduce variability between individual clones within the one-way ANOVA. Means ± 1 S.E. Multiple comparisons between means of treatment groups based on Tukey's HSD test. Due to rodent damage, only results of mineralized nitrogen from undamaged incubation bags are presented ($n = 16$). Root biomass based on a subset of clones ($n = 12$).

		Area			F	df	p
		Center	Middle	Fringe			
Thatch	Depth (cm)	68.8 $\pm$ 3.5 ^a	85.4 $\pm$ 4.1 ^b	53.2 $\pm$ 3.7 ^c	26.9	2, 66	≤ 0.0001
	Mass (g/m ²)	124.7 $\pm$ 10.4 ^a	160.3 $\pm$ 14.9 ^a	125.3 $\pm$ 12.0 ^a	3.2	2, 66	0.0481
Litter	Depth (cm)	14.1 $\pm$ 1.0 ^{ab}	15.2 $\pm$ 1.1 ^a	10.7 $\pm$ 0.7 ^b	4.3	2, 66	0.0182
	Mass (g/m ²)	175.2 $\pm$ 17.3 ^{ab}	198.5 $\pm$ 19.9 ^a	135.0 $\pm$ 16.9 ^b	3.5	2, 66	0.0374
Root mass (g)	0–10 cm	12.5 $\pm$ 1.3 ^a	12.7 $\pm$ 1.3 ^a	8.9 $\pm$ 0.6 ^a	4.0	2, 22	0.0323
	10–30 cm	2.5 $\pm$ 0.3	4.0 $\pm$ 0.7	3.4 $\pm$ 0.7	1.7	2, 22	0.2054
Moisture (%)	August 2004	5.2 $\pm$ 0.5 ^a	4.4 $\pm$ 0.3 ^{ab}	3.6 $\pm$ 0.2 ^b	6.7	2, 66	0.0022
	July 2005	9.5 $\pm$ 0.8 ^a	8.5 $\pm$ 0.6 ^{ab}	7.5 $\pm$ 0.6 ^b	4.1	2, 66	0.0206
Available N (mg/kg) ^a	July 2004	19.20 $\pm$ 1.71	21.25 $\pm$ 2.07	19.90 $\pm$ 2.44	1.3	2, 66	0.2816
	July 2005	33.40 $\pm$ 2.61 ^a	28.79 $\pm$ 2.16 ^{ab}	27.12 $\pm$ 1.85 ^b	7.3	2, 66	0.0014
Mineralized N (mg/kg) ^a	July–Aug 2004	9.03 $\pm$ 1.57	14.98 $\pm$ 2.65	10.77 $\pm$ 3.38	1.8	2, 30	0.1912
Olsen P (mg/kg)		0.99 $\pm$ 0.09 ^a	0.64 $\pm$ 0.06 ^b	0.66 $\pm$ 0.05 ^b	16.8	2, 66	≤ 0.0001

^a [NH₄⁺ N] + [NO₃⁻ N].

root crown, dried (48 h, 60 °C) and weighed them, and replanted new seedlings into the same pot.

2.4. Data analysis

We used one-way analysis of variance (ANOVA) to compare the density, aboveground biomass, and height of smooth brome in the center, middle, and along the fringe of invading clones. Means of treatment groups were blocked by clone ($n = 34$) and, where required, data were log transformed to meet model assumptions (Legendre and Legendre, 1998). We also used ANOVA to compare the abundance of plant parasitic nematodes in soil extracted from clones of smooth brome. Impacts of soil origin, pasteurization, and growth cycle on the biomass of smooth brome seedlings were tested using a three-way ANOVA. Data were blocked to account for variance in the biomass of seedlings raised in the greenhouse among selected clones ($n = 6$) and were log transformed to meet model assumptions (Legendre and Legendre, 1998).

Physical environments (available nitrogen, phosphorus, soil moisture) in the center, middle, and fringe of invading clones were also compared using one-way ANOVA. Where required, data were log or arcsine transformed to meet model assumptions (Legendre

and Legendre, 1998). Statistical analyses were performed using R version 3.1.2 (R Core Team, 2014).

We calculated indices of community diversity and equitability to examine the impacts of smooth brome invasions on the diversity of northern fescue prairies. Effective richness [$N_2 = (\sum p_i^2)^{-1}$] and evenness [$E3 = [(\sum p_i^2)^{-1} - 1] / [\exp(-\sum p_i \log p_i) - 1]$] describe communities based on the proportional abundance of species and the shape of their frequency distributions and can be used to examine the intensity of interspecific interactions (Legendre and Legendre, 1998).

3. Results

Ramets of smooth brome were less vigorous inside the invading clones. Although more dense ($F_{2,66} = 12.2$, $p \leq 0.0001$; Fig. 1A.), biomass of individual ramets at the center of invading clones declined compared to those along the invading fringes ($F_{2,66} = 19.1$, $p \leq 0.0001$; Fig. 1B). Despite the higher density of ramets at the center of invading clones, we observed no significant change in the height ($F_{2,66} = 2.9$, $p = 0.0591$) or the productivity of smooth brome ($F_{2,66} = 3.1$, $p = 0.0537$) among different areas of invading clones (Fig. 1C and D, respectively). Invading clones were also

Table 2

Mean number of plant parasitic nematodes (kg⁻¹ dry soil) extracted from the soil of twelve clones of smooth brome (*Bromus inermis* Leyss.) invading two northern fescue prairies in Riding Mountain National Park, Manitoba, Canada. The areas, center, middle, and fringe, represent a transect from the center of invading clones to their prairie fringes. Nomenclature according to Mai and Mullin (1996). Feeding habits according to Yeates et al. (1993). Means ± 1 S.E. Numbers in brackets refer to single sample occurrences.

		Area			F	df	p
Genus	Feeding Habit	Center	Middle	Fringe			
<i>Criconebella</i>	Ectoparasite	308 $\pm$ 103	422 $\pm$ 76	321 $\pm$ 28	0.6	2, 22	0.5678
<i>Helicotylenchus</i>	Semi-endoparasite	577 $\pm$ 173	629 $\pm$ 209	529 $\pm$ 212	0.1	2, 60	0.9388
<i>Hemicriconebellodes</i>	Ectoparasite	725 $\pm$ 401	347 $\pm$ 110	294 $\pm$ 95	1.3	2, 19	0.2894
<i>Hemicyclophora</i>	Ectoparasite	615 $\pm$ 296	577 $\pm$ 268	1448 $\pm$ 694	1.1	2, 17	0.3576
<i>Heterodera</i>	Sedentary endoparasite	147 (1)	135 $\pm$ 75	174 $\pm$ 88	0.1	2, 6	0.9471
<i>Hoplolaimus</i>	Semi-endoparasite	130 $\pm$ 37	131 $\pm$ 24	328 $\pm$ 110	3.6	2, 8	0.0771
<i>Paratylenchus</i>	Ectoparasite	103 $\pm$ 95	112 $\pm$ 105	324 $\pm$ 1	2.4	2, 3	0.2425
<i>Pratylenchus</i>	Migratory endoparasite	–	151 $\pm$ 48	165 $\pm$ 105	0.0	1, 2	0.9425
Unidentified (juvenile)		33 (1)	40 $\pm$ 34	35 $\pm$ 25	0.0	2, 6	0.9901
	Sum	2637 $\pm$ 1104	2545 $\pm$ 1410	3618 $\pm$ 950	0.1	2, 166	0.9060

characterized by higher depth and biomass of thatch and litter, and higher biomass of shallow roots inside the clone (Table 1). Soil moisture and available nitrogen and phosphorus increased in the center of invading clones, however, rates of nitrogen mineralization remained unchanged (Table 1).

Decreased vigor of smooth brome at the center of invading clones was not associated with plant parasitic nematodes. The mean number of plant parasitic nematodes was not significantly different between the center, middle, and the fringes of invading clones ($F_{2,166}=0.1$, $p=0.9060$; Table 2). With the exception of species of *Criconebella* and *Helicotylenchus*, which were most abundant in the middle of invading clones, and *Hemicriconebellodes*, which were most abundant in the center, the abundance of plant

parasitic nematodes was highest along the fringes of invading clones (Table 2). However, these differences were not significant (Table 2).

Soilborne pathogens did not have a significant impact on the biomass of smooth brome seedlings raised in soil collected from the center of invading clones. The effect of soil pasteurization decreased with time and, in the absence of pasteurization, the biomass of seedlings raised in soil collected at the center of invading clones continued to increase with each consecutive growth cycle ($F_{4,95}=119.5$, $p\leq 0.0001$; Fig. 2A). Although the biomass of seedlings raised in soil collected in prairies adjacent to invading clones was higher during most growth cycles ($F_{4,95}=6.2$, $p=0.0002$; Fig. 2B), the lower biomass of seedlings raised in soil

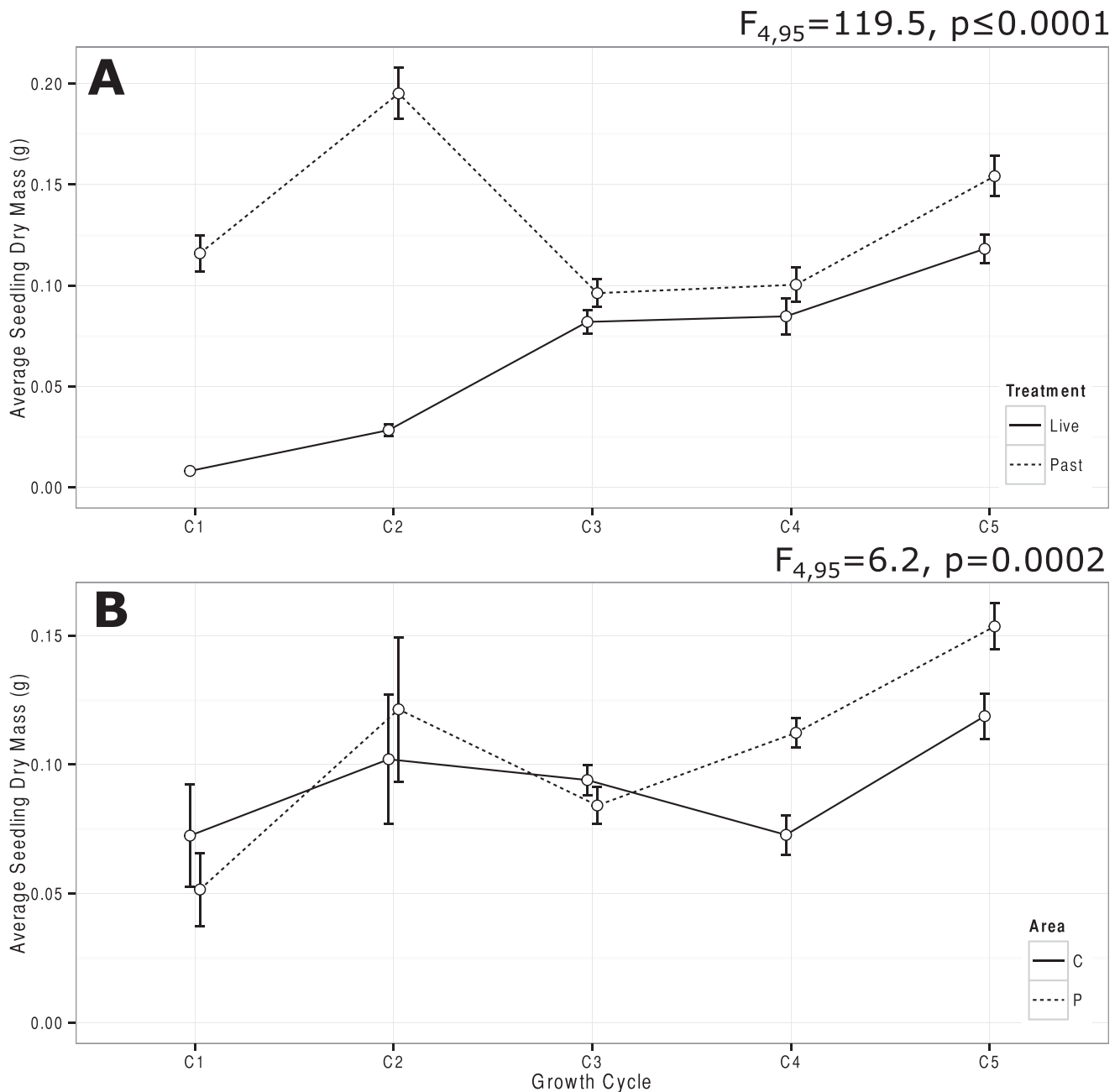


Fig. 2. (A) Aboveground dry biomass (g) of smooth brome (*Bromus inermis* Leyss.) seedlings raised in soil collected from the center and native prairie adjacent to clones invading northern fescue prairies in Riding Mountain National Park, Manitoba, Canada. (A) Biomass of seedlings raised in unpasteurized (solid lines) and pasteurized soil (dashed line). (B) Biomass of seedlings raised in soil collected from the center of invading clones (solid lines) and adjacent native prairie (dashed lines). Seedlings were raised in the greenhouse and harvested at the end of five consecutive five-week cycles. Means $\pm$ 1 S.E.

Table 3

Effects of soil pasteurization, area (prairie vs. center), and growth cycle (1–5) on the mean aboveground dry biomass of seedlings of smooth brome (*Bromus inermis*) raised in the greenhouse. Seedlings were germinated from seeds of smooth brome invading northern fescue prairies in Riding Mountain National Park, Manitoba, Canada. ANOVA models were blocked by clone ($n = 6$). Values of seedling biomass were log transformed.

Source	df	SS	F	P
Pasteurize	1	32.81	473.3	≤ 0.0001
Area	1	0.57	8.3	0.0050
Cycle	4	29.29	105.6	≤ 0.0001
Clone	5	0.53	1.5	0.1919
Pasteurize $\times$ area	1	0.01	0.1	0.7035
Pasteurize $\times$ cycle	4	33.14	119.5	≤ 0.0001
Area $\times$ cycle	4	1.73	6.2	0.0002
Pasteurize $\times$ area $\times$ cycle	4	0.61	2.2	0.0769
Residuals	95	6.59		

collected at the center of invading clones was characteristic of both pasteurized and unpasteurized soils ($F_{1,95} = 0.1$, $p = 0.7035$; Table 3). As a result, soil pasteurization did not affect the growth of plants raised in soil from the center of clones, but not from prairie soil either, regardless of time ($F_{4,95} = 2.2$, $p = 0.0769$; Table 3).

Despite the decreased vigor of smooth brome at the center of invading clones, plant communities inside the clones were characterized by a lower cover of native species ($F_{3,99} = 89.5$, $p \leq 0.0001$) and lower plant species richness ($F_{3,99} = 23.3$, $p \leq 0.0001$; Table 4) compared to the adjacent native prairies. Higher cover of smooth brome and lower plant community evenness were also characteristic of areas inside the clone ($F_{3,66} = 39.7$, $p \leq 0.0001$ and $F_{3,66} = 10.0$, $p \leq 0.0001$, respectively; Table 4).

4. Discussion

The vigor of smooth brome invading northern fescue prairies declined in the centers of clones. The observed loss of vigor in this area was not attributable to the number of plant parasitic nematodes or soilborne pathogens. Fewer soilborne plant parasitic nematodes were found in the older centers of invading clones and the mass of smooth brome seedlings raised in soil collected from the center of invading clones was not affected by pasteurization.

The observed decline in the vigor of smooth brome is characteristic of other clonal species. Watt (1947) first reported a ring-like morphology in tussocks of sheep fescue (*Festuca ovina* L.) and bent-grass (*Agrostis tenuis* Sibthorp), and Weaver (1968) described deterioration of plants at the center of clumps of big bluestem (*Adropogon gerardii* Vitman.). Stands of purple small reed (*Calamagrostis canescens* (G.H.Weber) Roth) growing in fens have also been reported to lose vigor and density and become replaced

by invasive forbs (Soukupova, 1992). The observed decline in the vigor of clonal species has been observed in arable fields with monocultures of annual plants and has been studied in other perennial plants, including *Jacobaea vulgaris* (van de Voorde et al., 2012).

Although decline in the vigor of clonal species has been attributed to a buildup of soilborne pathogens (de Rooij-van Der Goes, 1995), invasive plants may benefit from a release from soilborne pathogens that regulate their populations in the native range (Agrawal et al., 2005). However, even though the early stages of their invasion may be characterized by an escape from soilborne pathogens, these may accumulate with time (Dietz et al., 2010) or as successful invaders expand their geographic range (Hawkes, 2007). Escape from pathogens across multiple guilds of enemies is rare (Agrawal et al., 2005) and it remains possible that the observed loss of vigor of smooth brome in the center of invading clones is attributable to other groups of pathogens, especially those with steep dispersal gradients. For example, the potential impact of splash dispersed rusts, mildews, and endophytes on the vigor of exotic invaders remains unexplored (Mitchell et al., 2002; Rudgers et al., 2005). Also possible is the residual effect of a historic community of soilborne pathogens, whose legacy is the observed loss of vigor of smooth brome at the center of invading clones (Harper, 1990). Despite our focus on soilborne pathogens, negative feedbacks in the rhizosphere following invasion, could also include changes in populations of host-specific rhizosphere bacteria and mycorrhizal fungi (Bever, 2003). These questions remain unanswered.

In the absence of a demonstrated negative plant-soil feedback on the loss of vigor of smooth brome clones, a number of hypotheses remain untested. Several authors have attributed the loss of vigor of clonal species to their senescence. For example, both Watt (1947) and Weaver (1968) attributed the ring-like growth of clonal grasses to the aging of ramets, and Tamm et al. (2002) reported higher rates of ramet turnover among perennial species in meadow communities. Similar observations of self-thinning among clonal species, including trembling aspen (*Populus tremuloides* Michx.; Krasny and Johnson, 1992) and Japanese knotweed (*Reynoutria japonica* Houtt.; Adachi et al., 1996), suggest potentially weak physiological integration among ramets that could increase intra-clonal competition and result in stand thinning (de Kroon, 1993). Based on our observations, potential senescence of ramets at the center of invading clones of smooth brome could be signaled by higher concentrations of soil nitrogen, phosphorus, and soil moisture in the center of clones. In addition, senescence of older ramets at the center of the clones could explain the observed decline in the abundance of soilborne plant parasitic nematodes.

The density of nematodes remains closely linked with the function of plant roots (Yeates and Coleman, 1982) and lower

Table 4

Impacts of smooth brome (*Bromus inermis* Leyss.) invasion on the plant diversity of two northern fescue prairies in Riding Mountain National Park, Manitoba, Canada. The four areas (center, middle, fringe, prairie) contrast plant community diversity inside invading clones and adjacent prairies and were compared using 1 m² quadrats. Means of treatment groups were blocked by clone ($n = 34$) to reduce variability between individual clones within the one-way ANOVA. Means $\pm$ 1 S.E. Multiple comparisons between means of treatment groups based on Tukey's HSD test.

	Area				F	df	p
	Center	Middle	Fringe	Prairie			
Total cover (%)							
% native species	56.7 $\pm$ 3.4 ^a	59.3 $\pm$ 3.6 ^a	80.0 $\pm$ 2.1 ^b	98.8 $\pm$ 0.5 ^c	89.5	3, 99	≤ 0.0001
% smooth brome	41.7 $\pm$ 3.2 ^a	39.7 $\pm$ 3.6 ^a	18.6 $\pm$ 2.0 ^b	–	39.7	2, 66	≤ 0.0001
Richness (S)	12.5 $\pm$ 0.7 ^a	13.4 $\pm$ 0.7 ^a	17.3 $\pm$ 0.6 ^b	17.4 $\pm$ 0.6 ^b	27.1	3, 99	≤ 0.0001
Shannon's index [$N1 = \exp(-\sum p_i \log p_i)$]	2.2 $\pm$ 0.1 ^a	2.3 $\pm$ 0.1 ^a	2.8 $\pm$ 0.1 ^b	2.8 $\pm$ 0.1 ^b	21.8	3, 99	≤ 0.0001
Effective richness [$N2 = (\sum p_i^2)^{-1}$]	4.6 $\pm$ 0.4 ^a	5.3 $\pm$ 0.4 ^a	7.7 $\pm$ 0.4 ^b	7.6 $\pm$ 0.5 ^b	15.4	3, 99	≤ 0.0001
Evenness [$E3 = (N2 - 1)/(N1 - 1)$]	2.7 $\pm$ 0.1 ^a	2.9 $\pm$ 0.2 ^a	3.6 $\pm$ 0.1 ^b	3.5 $\pm$ 0.2 ^b	10.0	3, 99	≤ 0.0001

numbers of soilborne plant parasitic nematodes at the center of invading clones may indicate a change in the function of smooth brome roots with clone age. Although we did not examine the abundance of nematodes in the middle of invading clones, it is possible that the effect of their suppression on the growth of smooth brome was stronger in this area. In cases of high densities of plant parasitic nematodes, host plants may die, leading to lower postharvest population densities of nematodes in the center of infested areas compared to those along vigorous perimeters (van der Stoep et al., 2002; Been and Schomaker, 2006). In our case, it is conceivable that soil collected from the center of invading clone may have had a less strong effect on the growth of smooth brome seedlings than soil collected from the middle because the numbers of nematodes and pathogens may have declined after die-back of stems in the center. In such a case, including soil from the middle of invading clones may have resulted in stronger effects. This hypothesis remains to be tested.

Equally important in the vigor of invading clones may be their impact on nutrient cycling. Depletion of nutrients has been proposed as a potential cause of the observed die-back at the center of clonal plants (Ehrenfeld et al., 2005). Although the immobilization of nitrogen in the litter of exotic invaders, including smooth brome, could lead to their gradual decline (Christian and Wilson, 1999), feedbacks between litter quality, soil environments, and soil microbial communities are likely key to the outcome of litter deposition (Facelli and Pickett, 1991; Cragg and Bardgett, 2001). For example, Vinton and Goergen (2006) attribute the vigor of smooth brome to the higher efficiency of N cycling under invading clones, and Wedin and Tilman (1990) report a similar increase in the mineralization of nitrogen from the nitrogen rich litter of Kentucky bluegrass (*Poa pratensis* L.) and quack-grass (*Agropyron repens* (L.) Beauv.). Our observations of the greater availability of nitrogen and phosphorus at the center of invading clones do not support the immobilization hypothesis, and we found no significant differences in the rates of nitrogen mineralization inside the invading clones. While the observed loss of vigor among smooth brome ramets and the increased availability of soil resources at the center of invading clones supports the senescence hypothesis, other mechanisms remain untested. For example, the observed accumulation of thatch, litter, and the formation of shallow root mats at the center of invading clones could reduce access to moisture and inhibit the uptake of nutrients by ramets, leading to their loss of vigor (Weaver and Rowland, 1952). Alternatively, litter accumulation could induce central die-back in clonal plants as a result of its autotoxic effects (Carteni et al., 2012). To our knowledge, these hypotheses remain untested.

4.1. Persistent impacts of an exotic invader

Despite the observed impacts of pathogens on species turnover in plant communities (van der Putten et al., 1993; Mangan et al., 2010), most studies of biological invasions fail to incorporate sufficiently long temporal scales to evaluate long-term changes in ecological communities that determine the persistence of exotic organisms (Hawkes, 2007). Our results, measuring the vigor of smooth brome clones from their centers to their invading fringes, illustrate that exotic invasive plants can eventually lose their initial vigor (Dietz et al., 2010). Despite the observed loss of vigor, the impacts of smooth brome on plant community diversity of northern fescue prairies were persistent, and the accumulating thatch and litter inside the invading clones are likely to continue to hinder future restoration of prairie communities (Weaver and Rowland, 1952). Restoration of invaded areas is also likely to remain complicated by the accumulation of seed and rhizome banks, and the potential chemical alteration of prairie soils (D'Antonio and Meyerson, 2002) that could facilitate the

establishment of other exotic species (Bonanomi et al., 2005; Jordan et al., 2007). Restoring areas invaded by smooth brome, despite the observed decline in the vigor of invading clones, will likely require active methods of restoration to help rejuvenate seedbanks of native species and increase the access of native plants to soil resources.

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